

Robotic Exam 17/06/2020

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You should only have a browser open and no other applications. The browser should have only the exam tabs open. The exam is saved into the browser cookies so if you close it and reopen it nothing should happen. This is unless you browse in incognito mode or you delete all cookies upon browser closure. Do it at your own risk.

Do not maximize the window of your browser.

When working on paper your camera should look at you working on the paper, when working on computer the camera should look at you. You cannot go out of sight. Your microphone should be on all the time.

In case a question require to upload a file, this should be a pdf file of, no more than 10MB, whose name follows this format (Files not satisfying these rules will be penalized):

- PERSONAL_ID (for instance: 10920987) NOT YOUR ENROLLMENT NUMBER (for instance: 973898))
- then _
- then the NUMBER OF THE QUESTION
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You will have 2 hours, after which the Form is automatically closed and there will no possibilities to submit your answers any more. You will be noticed 15' in advance in case you have not uploaded files you should take care of that. Exams which are not submitted withing the given time will be considered as RITIRATO.

In case you do not see the IMAGES try to reload the page. Do it immediately, and in case it does not work use the following link

<https://www.dropbox.com/s/ifcl7f2wwtjg18z/Robotics20200617Images.pdf?dl=0>
(<https://www.dropbox.com/s/ifcl7f2wwtjg18z/Robotics20200617Images.pdf?dl=0>).

Punti:0/27

Section

EXERCISE 1: Robot Kinematics

Consider the robot in the picture and the associated equations.



$$v = \frac{V_r + V_l}{2}$$

$$\omega = \frac{V_r - V_l}{b}$$

1

What are the formulas about? 
 (-/1 punti)

2

What does b refer to in the formulas? What happens if you measure it longer than it actually is in the real robot? 
 (-/2 punti)

3

What happens if the robot has the right tyre more inflated than the left one?  
(-/1 punti)

4

What is the ICC? Where the ICC of this robot can be? Are there special locations?  
(-/2 punti)

Section

EXERCISE 2: Localization

In the following we assume we want to localize a robot dog in an arena surrounded by 6 colored posts in known positions. One of the possible implementation of the Bayes filter for localization is the Kalman Filter (KF) or its non linear version known as the Extended Kalman Filter (EKF).



5

Write the sensor measurement model in this case and discuss it in details. Be practical in the answer as you should be ready to implement it yourself. 🗨️
(-/2 punti)

6

Do we need an Extended Kalman filter or a Kalman filter is enough? Why? 🗨️
(-/1 punti)

7

Build the Extended Kalman Filter (EKF) selecting the correct steps (NOTE: Multiple choices apply) 
(-/3 punti)

- ☐ $\mu' = g(\mu, u)$
- ☐ $\mu' = G * \mu$
- ☐ $\mu' = G * \mu + B * u$
- ☐ $\Sigma' = G^T * \Sigma * G + R$
- ☒ $K = \Sigma * H^T * (H * \Sigma' * H^T + Q)^{-1}$
- ☐ $K = \Sigma' * C^T * (C * \Sigma' * C^T + Q)^{-1}$
- ☒ $\mu = \mu' + K(z - h(\mu'))$
- ☐ $\mu = \mu' + K(z - H * \mu')$
- ☒ $\Sigma = (I - K * H) * \Sigma'$
- ☐ $\Sigma = (I - K * C) * \Sigma'$

8

What are H and G in the previous question? 
(-/1 punti)

EXERCISE 3: Simultaneous Localization and Mapping

Let assume we do NOT know the map of colored posts, i.e., corner poles, in the RoboCup scenario in the figure. What we know is how many posts can be there and their special color codes.



9

Design a SLAM system for this scenario. Provide the elements you would need to implement it. 🗉
(-/2 punti)

10

For this problem, it is better to use Online SLAM or a Full SLAM? Why? 
(-/1 punti)

Exercise 4: Robot Operating System

Let's assume we want to build an exam proctoring system based on several cameras monitoring students doing their robotics exam at home.

11

Could we use ROS to implement it? How? ... I just need the rationale behind your answer here. 
(-/2 punti)

12

What would you use services for in your system? Make an example! 
(-/2 punti)

Section

EXERCIZE 5: Robot Motion Control

Let's consider a sanitizing wheeled robot robot which each night has to traverse a university campus spraying around disinfectant to sanitize the areas before the students can access them.



13

What are the characteristics of your robot in terms of kinematics and payload? Why?


(-/2 punti)

14

Which of the obstacle avoidance algorithms would you chose among VHF, VHF+, and DWA for your robot? Why? 
(-/2 punti)

15

Let assume we want to control a drone in an indoor environment. How would you implement its obstacle avoidance system? Why? 
(0/2 punti)



Inserisci la risposta

16

Is a drone a good idea to solve the problem posed in this exercise? Why? 
(-/1 punti)

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